

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

SEWAGE TREATMENT PLANT (STP)

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. Plant Profiling & Overview

A Sewage Treatment Plant (STP) treats domestic/municipal-strength wastewater generated from residential complexes, townships, hotels, hospitals, commercial buildings, and industrial colonies before disposal or reuse. Unlike an ETP, sewage is relatively uniform in character (mainly organic, biodegradable load) and does not typically contain industry-specific toxic pollutants, allowing standardized, compact biological technologies.

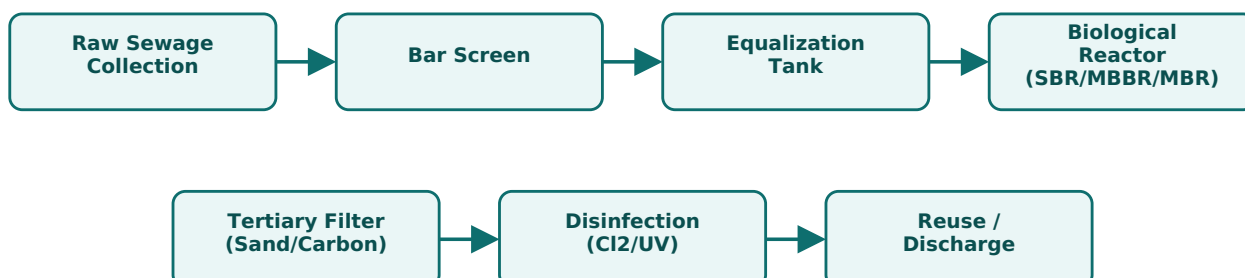


Fig. 1 — STP Process Train (SBR-based, typical for townships/buildings)

1.1 Objectives

- Reduce BOD, COD, TSS, and pathogens to levels safe for discharge or reuse.
- Enable treated water reuse for flushing, gardening, and HVAC cooling makeup (mandatory for buildings above notified plot-size/discharge thresholds in many municipal bye-laws).
- Prevent groundwater and surface water contamination from untreated sewage.
- Comply with local municipal corporation and State Pollution Control Board norms.

1.2 Typical Technologies Used

- Sequential Batch Reactor (SBR) – most common for townships/buildings; compact, automated.
- Moving Bed Biofilm Reactor (MBBR) – compact retrofit-friendly option, robust to load swings.
- Membrane Bioreactor (MBR) – premium reuse-grade quality, used where treated water is reused for flushing/HVAC within the same premises.
- Extended Aeration Activated Sludge – simple, low sludge production, larger footprint, common for larger municipal STPs.
- Fluidized Aerobic Bio Reactor (FAB) – biofilm-based, compact, tolerant to intermittent flows.

1.3 Regulatory Framework

- Consent to Establish/Operate from State Pollution Control Board for STPs above notified capacity.
- Municipal building bye-laws mandating STP and dual-plumbing reuse for group housing societies.
- CPCB general standards for discharge into inland surface water/land for irrigation/sewers.
- National Green Tribunal (NGT) directives mandating 100% treated sewage reuse in several states.

2. Sewage Characterization & Design Flow Basis

Design flow for residential/commercial STPs is estimated from per-capita water supply norms (typically 135-150 LPCD for residential, 45-70 LPCD for offices/institutions) with 80% return factor to sewage. Peak factor of 2.5-3x average flow is applied for hydraulic design of screens and equalization.

Parameter	Typical Raw Sewage Range	Typical Treated (Reuse Grade)
BOD	150-300 mg/L	≤ 10 mg/L
COD	300-600 mg/L	≤ 50 mg/L
TSS	150-350 mg/L	≤ 10 mg/L
Ammoniacal Nitrogen	20-40 mg/L	≤ 5-10 mg/L
Fecal Coliform	10 ⁶ -10 ⁸ MPN/100mL	≤ 100-230 MPN/100mL (post disinfection)
pH	6.5-8.5	6.5-8.5

3. Process Design & Engineering Basis

3.1 Design Parameters

- Hydraulic Retention Time: equalization 2-4 hr; SBR react-settle-decant cycle 4-6 hr per batch; extended aeration 18-24 hr.
- F/M ratio 0.05-0.15 kg BOD/kg MLSS.day (extended aeration, lower than industrial ETP).
- MLSS 3000-5000 mg/L (ASP/SBR); 4000-6000 mg/L carrier-supported biomass in MBBR.
- Sludge age 15-25 days for extended aeration (low sludge yield, infrequent desludging).
- Air requirement: 40-60 m³ air per kg BOD removed (diffused aeration).

3.2 Core Unit Operations

a) Bar Screen & Equalization: Removes rags/solids; equalization tank buffers diurnal peak flows typical of residential usage patterns (morning/evening peaks).

b) Biological Reactor (SBR/MBBR/MBR): Aerobic degradation of organics with programmable fill-react-settle-decant sequencing (SBR) or fixed-film carriers (MBBR) or membrane-coupled biomass separation (MBR) for superior effluent clarity.

c) Tertiary Filtration & Disinfection: Pressure sand filter → activated carbon filter (odor/color polishing) → chlorination/UV disinfection to meet fecal coliform limits before reuse.

4. Pretreatment Systems

Stage	Purpose	Typical Equipment
Bar Screen	Remove rags, sanitary items, plastics	Manual/mechanical raked screen, 15-25mm spacing
Oil & Grease Trap	Remove kitchen/food-court grease	Grease trap ahead of common collection sump
Equalization Tank	Buffer diurnal peak flows	RCC tank with diffused aeration/mixing
Grit Removal (large STPs)	Remove sand/silt ingress	Vortex grit chamber

5. Secondary & Tertiary Treatment

Secondary biological treatment (SBR/MBBR/MBR/extended aeration) reduces BOD by 90-95%. Tertiary treatment – pressure sand filtration, activated carbon filtration, and disinfection (chlorination or UV) – polishes the effluent for horticulture, flushing, or cooling tower makeup reuse. Where treated water is used for cooling tower makeup, additional softening/RO may be needed to control scaling (refer to the Cooling Tower and Softener manuals).

6. Sludge Handling & Disposal

- Excess biological sludge is low in volume compared to ETP due to low F/M design; typically wasted every few days to a sludge holding tank.
- Dewatering via decanter centrifuge, screw press, or sludge drying beds for small/medium STPs.
- Dewatered cake (non-hazardous, nutrient-rich) can be used as manure/soil conditioner subject to local guidelines, or disposed to municipal solid waste processing facility.

7. Sewage & Disinfection Chemistry Deep-Dive

7.1 Aerobic Biodegradation Stoichiometry

Simplified aerobic oxidation: $C(a)H(b)O(c)N(d)$ (organics) + O_2 $\rightarrow$ biomass $\rightarrow$ CO_2 + H_2O + NH_3 + new cells
Nitrification (if aerobic SRT is long enough): $NH_4^+ + 2O_2 \rightarrow NO_3^- + H_2O + 2H^+$ (via Nitrosomonas/Nitrobacter)

Ammoniacal nitrogen in raw sewage (20-40 mg/L) is only oxidized to nitrate if sludge age and DO are sufficient for nitrifying bacteria to establish (nitrifiers grow much more slowly than heterotrophs) – this is why extended aeration/high-SRT designs achieve better ammonia removal than short-SRT conventional ASP, and why a sudden drop in sludge age (excess desludging) can cause ammonia breakthrough before BOD is affected.

7.2 Chlorination Disinfection Kinetics

Chick-Watson law: $\log(N/N_0) = -k \times C \times t$
where N/N_0 = surviving fraction, C = disinfectant residual concentration, t = contact time, k = die-off rate constant

Practical CT target for fecal coliform reduction to reuse-grade: residual chlorine 0.5-1.0 mg/L $\times$ 20-30 min contact time

Disinfection efficiency depends on the CT product, not concentration alone – a contact tank that is too small (short-circuiting) will under-perform even with correct chlorine dose. Turbidity above ~5-10 NTU shields bacteria from disinfectant contact, which is why tertiary filtration ahead of disinfection is essential to meeting the 100-230 MPN/100mL reuse target reliably.

7.3 UV Disinfection as an Alternative

UV disinfection inactivates pathogens by damaging DNA/RNA (thymine dimer formation) rather than by a chemical residual, so it produces no disinfection by-products (e.g., trihalomethanes) and requires no de-chlorination step before discharge to sensitive water bodies. UV dose (mJ/cm²) is a function of lamp intensity, flow rate and water transmittance (turbidity/color) – sleeve fouling and lamp aging are the main causes of dose shortfall in service (Section 8/10).

7.4 Detergent & Foam Chemistry

Surfactants from laundry/kitchen sources reduce the surface tension of mixed liquor, stabilizing air bubbles into a persistent foam layer, especially at high sludge age where slow-growing Nocardia-type filamentous organisms (which have a waxy, hydrophobic cell wall) accumulate at the air-water interface. Reducing sludge age and physically skimming/wasting the foam layer are the standard first responses; anti-foam chemical addition is a symptom-level fix, not a root-cause solution.

8. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
Odor near STP/society	Anaerobic equalization tank, inadequate aeration	Increase air supply, cover tanks, install biofilter
Frothing/foaming in aeration tank	Detergents from laundry/kitchen discharge, high sludge age	Skim foam, reduce sludge age, dilute detergent-heavy batches
Poor disinfection (high coliform in treated water)	Under-dosed chlorine, high turbidity shielding pathogens, contact time too short	Increase chlorine dose/contact tank size, improve upstream filtration
SBR decanter jamming/blockage	Floating solids, rag ingress from inadequate screening	Improve screening, inspect/clean decanter mechanism regularly
Effluent reuse water causing scaling in cooling towers	High hardness/TDS carried through STP	Add softening/RO polishing before reuse for cooling makeup
Sudden low BOD removal after holiday/low-occupancy period	Starved biomass due to low organic load	Maintain minimum recirculation, consider load-shedding aeration control
High MLSS carryover to tertiary filters	Poor settling, high SVI, clarifier/settling zone overload	Reduce MLSS, check polymer dosing if used, inspect decanter/weir levels

9. Equipment Service Life

Equipment	Typical Service Life	Key Life-limiting Factors
RCC/Fiber Tanks	20-30 years (RCC); 12-15 years (FRP)	Waterproofing, UV degradation (FRP), structural loading
Air Blowers	10-15 years	Bearing wear, filter maintenance, duty cycling
Fine Bubble Diffusers	5-7 years	Fouling, membrane stiffening
Submersible Sewage Pumps	5-8 years	Rag/solids clogging, seal wear, corrosion
MBBR Media	15-20 years	Mechanical abrasion
UV Disinfection Lamps	9,000-12,000 hours (~1-1.5 yrs continuous)	UV intensity decay, sleeve fouling
Decanter/Screw Press	10-12 years	Abrasive wear, bearing/seal wear
Instrumentation (DO/pH/level)	2-5 years	Fouling, calibration drift

10. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Check screen chamber, remove rags; monitor DO/MLSS visually; check chlorine dosing tank level; log flow meter
Weekly	MLSS/SVI testing; clean diffusers if surfacing uneven; inspect blower belts; grease trap cleaning
Monthly	Lab test of treated water (BOD, COD, TSS, coliform); calibrate online instruments; inspect UV sleeves
Quarterly	Vibration check on blowers/pumps; inspect civil tanks for cracks; sludge disposal compliance review
Annual	Complete internal inspection of all tanks; blower/pump overhaul; replace UV lamps; statutory audit and CTO renewal check

11. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Treated water still has odor	Insufficient aeration/detention, anaerobic pockets	Increase air supply, check for dead zones/short-circuiting
High coliform count despite chlorination	Contact tank short-circuiting, chlorine demand not met	Check baffling in contact tank, increase dose, verify residual chlorine
SBR not decanting fully	Decanter arm jam, level sensor fault	Inspect and clean decanter, check float/level switches
Sludge not settling (SBR/clarifier)	Bulking, low DO, overloaded biomass	Check SVI, adjust aeration and sludge wasting rate
Low treated water flow to reuse tank	Filter clogging, pump wear	Backwash sand/carbon filter, inspect pump impeller
High power consumption	Continuous blower run without DO control	Install DO-based blower cycling/VFD

12. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Feasibility & Load Estimation	Population/occupancy-based flow estimation, site survey, technology selection, budgetary cost.
Basic Engineering	PFD, mass balance, preliminary layout, equipment sizing, capex estimate.
Detailed Engineering	P&ID, GA drawings, structural/civil design, electrical SLD, control philosophy, tender BOQ.
Statutory Approvals	CTE from SPCB, municipal building plan STP approval, fire NOC.
Procurement	Blower, pump, SBR/MBBR internals, MCC/PLC panel, UV system procurement and inspection.
Civil & Structural	Underground/above-ground RCC tank construction, piping, plant room civil works.
MEI Erection	Equipment installation, piping, cabling, panel wiring, SCADA/PLC integration.
Pre-commissioning	Dry run, hydro test of tanks, loop checking, safety interlocks.
Commissioning & Biomass Seeding	Wet trial, seed sludge inoculation, 2-4 week acclimatization.
Performance Guarantee Test Run	Continuous run (72 hrs-30 days) verifying outlet quality at design flow.

13. Statutory Compliance & Performance Guarantee

Parameter	Typical Reuse-Grade Limit
BOD	≤ 10 mg/L
COD	≤ 50 mg/L
TSS	≤ 10 mg/L
Fecal Coliform	≤ 100-230 MPN/100mL
Residual Chlorine	0.5-1.0 mg/L (post disinfection)

Successful Performance Guarantee Test Run (PGTR), typically witnessed and lab-verified over a continuous 72-hour to 30-day run at design flow, triggers final payment and initiates the Defect Liability Period (commonly 12 months).